PRACTICAL- 5

AIM

Write a program to implement Tic-Tac-Toe game problem

THEORY

Tic-Tac-Toe is a simple two-player strategy game played on a 3×3 grid. The players take turns placing their symbols on the board. One player uses the symbol X and the other uses O. The objective of the game is to place three of the same symbols in a row either horizontally, vertically, or diagonally.

In this program, the Tic-Tac-Toe game is implemented using Python programming. The game board is represented using a two-dimensional list (3×3 matrix). Each cell in the matrix represents a position on the board where a player can place their symbol.

The program allows a human player and the computer to play the game. The user enters the row and column position where they want to place their symbol. After each move, the program checks whether a player has won the game. This is done by checking all rows, columns, and diagonals of the board.

If a player successfully places three symbols in a line, that player is declared the winner. If all the cells on the board are filled and no player has achieved three symbols in a row, the game ends in a draw.

This program demonstrates the use of Python concepts such as lists, loops, conditional statements, and functions to create a simple interactive game.

SOFTWARE USED

Jupyter Notebook

CODE

```
def print_board(board):
    for row in board:
        print(" | ".join(row))
        print("-" * 9)



def check_winner(board, player):
```

```python
    for i in range(3):
        if all(board[i][j] == player for j in range(3)):
            return True
        if all(board[j][i] == player for j in range(3)):
            return True

    if board[0][0] == board[1][1] == board[2][2] == player:
        return True

    if board[0][2] == board[1][1] == board[2][0] == player:
        return True

    return False


def board_full(board):
    for row in board:
        if " " in row:
            return False
    return True


board = [[" " for _ in range(3)] for _ in range(3)]

current_player = "X"

while True:

    print_board(board)
```

```
if current_player == "X":
    row = int(input("Enter row (0-2): "))
    col = int(input("Enter column (0-2): "))
else:
    print("Computer's Move")
    for i in range(3):
        for j in range(3):
            if board[i][j] == " ":
                row, col = i, j
                break
        else:
            continue
        break

if board[row][col] == " ":
    board[row][col] = current_player
else:
    print("Invalid Move")
    continue

if check_winner(board, current_player):
    print_board(board)
    print("Player", current_player, "wins!")
    break

if board_full(board):
    print_board(board)
    print("Game Draw")
```

```
            break


        current_player = "O" if current_player == "X" else "X"
```

OUTPUT

```
  |   |
---------
  |   |
---------
  |   |
---------
Enter row (0-2):  1
Enter column (0-2):  1
  |   |
---------
  | X |
---------
  |   |
---------
Computer's Move
O |   |
---------
  | X |
---------
  |   |
---------
Enter row (0-2):  2
Enter column (0-2):  1
O |   |
---------
  | X |
---------
  | X |
---------
Computer's Move
O | O |
---------
  | X |
---------
  | X |
---------
Enter row (0-2):  2
Enter column (0-2):  0
O | O |
---------
  | X |
---------
X | X |
---------
Computer's Move
O | O | O
---------
  | X |
---------
X | X |
---------
Player O wins!
```